

Organizational Trust Quotient

Speaker Notes — Capstone Presentation

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Presenter split: Both — slides 1 & 16 · Vernon — slides 2–6 and 14–15 · Srilatha — slides 7–13

Plain-language talking points — a short numbered list per slide, restarting at 1 each slide. These match the notes embedded in the PowerPoint file.

Slide 1 · Reading team health from everyday messages

Speaker: **Vernon & Srilatha**

1. Welcome to all and thank you for attending our capstone presentation. Please notice the QR code on each screen. Please use it to find out more information about our project —I'm Vernon Cox, Srilatha Alla and I are the researchers for Organizational Trust Quotient capstone project.
2. We built the Organizational Trust Quotient, or OTQ.
3. The idea is to digitize the day-to-day team messages to understand how the team is doing.
4. We don't survey anyone; we use patterns that are already within the messages.
5. Today we will cover the problem, what we measure, the data, and the working tool we built.

Slide 2 · Preview - What we'll cover

Speaker: **Vernon**

1. Here's the quick roadmap for the next ten minutes.
2. First, the business problem — teamwork problems negatively effects the bottom line.
3. We discuss what OTQ looks at, and the data behind it.
4. Then the results and the portal we built.
5. Finally, the takeaways and where we'd go next.

Slide 3 · The Business Problem

Speaker: **Vernon**

1. Let's start with why this matters in dollars.
2. Unfocused teams where everyone's busy but pulling in different directions.
3. Disjointed teams where work happens in silos; things get duplicated or dropped between people.
4. Unheard teams where concerns surface too late, after they've become real problems.
5. All of these issues will eventually show on a balance sheet, burn hours, slow projects, and push good people out of your company.
6. That's the real cost of not understanding your team and many leaders don't see it until it's too late.

Slide 4 · The Problem - Trust is hard to see**Speaker: Vernon**

1. How do some companies check on this today? Mostly surveys — annual, quarterly, or on other timelines.
2. People tend to answer how they think they're supposed to.
3. Filling out surveys pull everyone away from real work to fill them out.
4. Meanwhile, teams send thousands of messages a day — with message signals that goes unused.
5. Our goal: an always-on team message interpretation system, with no extra work for anyone.

Slide 5 · What OTQ Looks At**Speaker: Vernon**

1. Here's the heart of it.
2. We measure engagement using the existing team messaging systems.
3. From that, we predict responsiveness and the levels of team interaction.
4. What we're curious about underneath is trust.
5. The honest part: we measure messaging and predict team interaction levels; trust is what we believe sits underneath.
6. We do NOT claim to measure trust directly.

Slide 6 · The Data - Where it comes from**Speaker: Vernon**

1. The data comes from any messaging tools that your office already uses.
2. Slack and Outlook for chats, Teams for meetings, and other messaging systems.
3. We combine them into machine useable data called events — for this project we used about 1,250 team events from the KJV bible .
4. This public team data was used to test our system.

Slide 7 · The Data - What's in it**Speaker: Srilatha**

1. Each row in our data is one interaction.
2. We are not using the actual message words.
3. We only keep simple details like who was involved, where it happened, how long it lasted, how many people joined, and a reply-speed clue.
4. So the focus is on counts and timing, not message content.
5. This keeps the approach privacy-friendly from the beginning.

Slide 8 · The Data - How we score it**Speaker: Srilatha**

1. Next, we turn each interaction into an engagement score.
2. A meeting gets more points than an email because it takes more time and attention.
3. Longer interactions and more people also increase the score.
4. Then we add everything by day to see which days were more active and which days were quieter.

Slide 9 · Under the Hood - The code

Speaker: **Srilatha**

1. This slide shows the actual code behind the scoring.
2. The code gives a base score based on the interaction type — meeting, chat, call, email, or other.
3. Then it adds points for duration and number of people.
4. After that, we use these daily scores as inputs to the model.
5. The main point is that our scoring rule is simple, transparent, and matches the code.

Slide 10 · The Data - A first look

Speaker: **Srilatha**

1. Before building the model, we looked at the data pattern.
2. Most interactions are small, usually between one or two people.
3. That makes sense because many workplace messages are quick conversations.
4. The larger group meetings are fewer, and they appear on the right side of the chart.
5. So, the data looks realistic before modeling.

Slide 11 · The Model

Speaker: **Srilatha**

1. The model learns the connection between engagement and reply speed.
2. We used Ridge Regression because our features are related to each other.
3. For example, a busy day may have more events, more people, and more total engagement.
4. Ridge helps keep the model stable and easy to explain.
5. We also tested it on unseen data and checked which features mattered most.

Slide 12 · What We Built

Speaker: **Srilatha**

1. Finally, this is not only a notebook.
2. We built a portal interface that can run the full process.
3. A user can upload or connect the data, and OTQ runs the scoring, model, and charts.

Slide 13 · A visual of the actual system

Speaker: **Srilatha**

1. This is OTQ interface we created for the end user.
2. When the Slack, Outlook, and TEAMS files are uploaded it will show the visual graphs you are seeing.
3. We are showing the important features of the OTQ interface here.
4. The end user can understand the communication level of the team members.
5. The result is day-by-day predictions that business users can understand without writing code.
6. Finally, our goal is for the OTQ to become a practical tool, not just a data science experiment.

Slide 14 · The Verdict

Speaker: **Vernon**

1. Here's the read in one view.
2. The top stage on each track, with a confidence score.
3. Plus the evidence: about 1,250 events, 184 people, over roughly six months.
4. So you get the headline and the proof together.
5. Again: an engagement-based indicator, not a clinical assessment — good for discussion, not a verdict to act on blindly.

Slide 15 · Take-Home Points

Speaker: **Vernon**

1. **Four things to remember.**
2. Always-on is better than surveying.
3. Understanding Team engagement allows for better teamwork.
4. We're honest about the limits — our focus is on measuring team messaging levels.
5. Private by design — Identities are swapped for anonymized IDs.

Slide 16 · What's Next

Speaker: **Vernon & Srilatha**

1. **Where we'd take this next.**
2. Run it on a real company's data.
3. Check our score against key performance indicators (KPI's) of the business [before and after installation].
4. That's how we move from a promising signal to a proven tool.

Thank you — happy to take questions.